JinWoo Lee

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Education

Georgia Institute of Technology | Atlanta, GA
Bachelor of Science in Electrical Engineering, GPA 4.0/4.0 May 2028

Arizona State University | Tempe, AZ August 2025 – December 2025
Completed 17 Credit Hours, GPA 4.3/4.3

Green River College | Auburn, WA June 2025
Associate of Science in Computer and Electrical Engineering
Highest Honors, Mathematics Division Award, Foundation Scholarships, GPA 4.0/4.0

Experience

Steven Black | Auburn, WA October 2024 – June 2025
Math Grader
- Graded weekly problem sets (5 problems) for three class sections (~60 students) in multivariable calculus and integral calculus, while communicating regularly with the instructor to ensure consistent grading policies and evaluation standards.

Adrienne Palmer | Auburn, WA September 2024 – June 2025
Math Tutor
- Tutored undergraduate students in calculus, linear algebra, differential equations and engineering physics, 15 hours a week.
- Explained problem-solving strategies and core concepts to support students' conceptual understanding.

Projects & Activities

FPGA-Based SoC Peripheral and Display Controller | RTL/FPGA Training Project June 2026 – July 2026
Designed and implemented Verilog RTL modules for peripheral interfaces, memory access, and display control on FPGA hardware.
- Designed a 64-byte register-mapped AMBA APB3 slave IP and verified read/write transactions using an APB master testbench.
- Implemented a custom PL-side BRAM controller in Verilog and debugged reset polarity, write timing, and read latency using ILA.
- Developed a 4-wire SPI OLED controller with command/data control, reset sequencing, display addressing, and verified it on board.

MOSFET and SONOS Device Simulation | Sentaurus TCAD Project May 2026
Simulated semiconductor device behavior and short-channel effects using Synopsys Sentaurus TCAD.
- Constructed and simulated MOSFET structures using Sentaurus SDE and SDevice, evaluating threshold voltage roll-off, subthreshold behavior, DIBL, and GIDL.
- Compared LDD and Halo+LDD structures to analyze leakage and electrostatic control and investigated SONOS memory operation.

MOSFET I-V Characterization | Semiconductor Device Personal Project January 2026 – April 2026
Built and characterized an n-channel MOSFET circuit to connect device physics with experimental measurements.
- Constructed a breadboard measurement setup and collected drain-current data under multiple gate and drain bias conditions.
- Estimated threshold voltage from measured data and analyzed cutoff, linear, saturation, and leakage behavior.

Digital Tram Control System Design | ASU Digital Design Project December 2025
Designed a digital tram control system using sequential logic to meet specified operational constraints.
- Optimized a 6-state system, achieving a compact implementation with only 3 D flip-flops, 5 AND, 3 OR, and 2 NOT gates.
- Verified functionality through digital simulation and systematic testing; achieved full score on the project.

Relevant Coursework

Circuit Analysis: AC/DC circuit analysis, frequency-domain analysis, op-amps, passive filters, Bode plots
Digital Design Fundamentals: Combinational and sequential logic, finite-state machines, Boolean logics, setup/hold timing analysis

Skills

Programming & Embedded: C, C++, Python, MATLAB, ESP32, Arduino, Vitis
Digital: Verilog HDL, RTL design, AXI4-Lite, APB3, SPI, UART, BRAM, Vivado, ILA, Quartus, Oscilloscope
Analog & Semiconductor: LTspice, PSpice/OrCAD, Cadence, TCAD device simulation
Languages: Korean (native), English (fluent)